Project: Worked examples
Title: EN 1993-1-1 member checks (basic)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet. $N_{pl,Rd} = A f_y/\gamma_{M0}$; $M_{c,Rd} = W f_y/\gamma_{M0}$ with $W = W_{pl}$ (class 1–2) or W_{el} (class 3); $V_{pl,Rd} = A_v f_y/(\sqrt{3}\,\gamma_{M0})$; $\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$ when $V_{Ed} > 0.5\,V_{pl,Rd}$ reduces the bending resistance by $(1 - \rho)$; $N_{cr} = \pi^2 EI/L_{cr}^2$, $\bar{\lambda} = \sqrt{(A f_y/N_{cr})}$, $\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$, $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$, $N_{b,Rd} = \chi A f_y/\gamma_{M1}$; cross-section: $N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$; member: $N_{Ed}/N_{b,Rd,min} + k_{yy} M_{y,Ed}/M_{y,Rd} + k_{zz} M_{z,Ed}/M_{z,Rd} \le 1$.

Ref: BS EN 1993-1-1 6.2.3 — Tension
Ref: BS EN 1993-1-1 6.2.4 — Compression
Ref: BS EN 1993-1-1 6.2.5 — Bending moment
Ref: BS EN 1993-1-1 6.2.6 — Shear
Ref: BS EN 1993-1-1 6.2.8 — Bending and shear
Ref: BS EN 1993-1-1 6.2.1(7) — Linear summation of utilisations
Ref: BS EN 1993-1-1 6.3.1.1, 6.3.1.2 — Uniform members in compression: buckling resistance, buckling curves
Ref: BS EN 1993-1-1 6.3.3 — Uniform members in bending and axial compression
Ref: Gardner & Nethercot, Designers' Guide to Eurocode 3 §6.3.1 — Imperfection factors for the buckling curves

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_Ed	Design axial force, TENSION positive and compression negative (the opposite sign to S6's N_Ed, which is compression positive; S2's N_e1 is tension positive)	-900	kN	reference: S2 frame sheet, column 1, ULS	-100000 … 100000 kN
M_y_Ed	Design moment about y-y (major)	40	kN*m	reference: S2 frame sheet, column 1, ULS	0 … 100000 kN*m
M_z_Ed	Design moment about z-z (minor)	0	kN*m	reference: S2 frame sheet	0 … 100000 kN*m
V_Ed	Design shear force	20	kN	reference: S2 frame sheet	0 … 100000 kN
f_y	Yield strength (for the thickness)	355	MPa	spec: BS EN 10025-2 S355, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
gamma_M0	Partial factor γ_M0 (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
gamma_M1	Partial factor γ_M1 (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
section_class	Cross-section class	1		assumption: UC 203×203×46 S355 is class 1 in compression (Blue Book).	any
A	Area A	58.7	cm^2	reference: S1 section sheet / Blue Book UC 203×203×46	1 … 100000 cm^2
W_pl_y	W_pl,y	497	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_el_y	W_el,y	450	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_pl_z	W_pl,z	231	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_el_z	W_el,z	152	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	4568	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	1548	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper)	203.2	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
b	Width b (helper)	203.6	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
t_w	Web thickness (helper)	7.2	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
t_f	Flange thickness (helper)	11	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
r	Root radius (helper)	10.2	mm	spec: BS EN 10365 UC 203×203×46	0 … 500 mm
eta	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
L_cr_y	Buckling length about y-y	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_cr_z	Buckling length about z-z	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
curve_y	Buckling curve about y-y	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, y-y	any
curve_z	Buckling curve about z-z	c		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, z-z	any
k_yy	Interaction factor k_yy (member check)	1		assumption: Simplified interaction: k_yy = 1.0.	0.5 … 2
k_zz	Interaction factor k_zz (member check)	1		assumption: Simplified interaction: k_zz = 1.0.	0.5 … 2

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. Lateral-torsional buckling is not checked: the compression flange is assumed restrained, or LTB is checked elsewhere. (6.3.2 not implemented.)
A3. The member interaction uses the user's k_yy and k_zz (default 1.0) with the smaller flexural-buckling reduction on the axial term; the Annex A/B interaction factors are not derived. (6.3.3 simplified.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. section_class (Cross-section class) is taken as 1. (UC 203×203×46 S355 is class 1 in compression (Blue Book).)
A6. L_cr_y (Buckling length about y-y) is taken as 4 m. (Pinned both ends: L_cr = L = 4 m.)
A7. L_cr_z (Buckling length about z-z) is taken as 4 m. (Pinned both ends: L_cr = L = 4 m.)
A8. k_yy (Interaction factor k_yy (member check)) is taken as 1. (Simplified interaction: k_yy = 1.0.)
A9. k_zz (Interaction factor k_zz (member check)) is taken as 1. (Simplified interaction: k_zz = 1.0.)

4. Working

[27] r_h = r
= 10.2 mm
= 10.2 mm
Root radius used by the helper

[28] A_v_rolled = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f 6.2.6(3)
= 58.7 cm² - 2 × 203.6 mm × 11 mm + (7.2 mm + 2 × 10.2 mm) × 11 mm
= 16.944 cm²
Rolled I shear area, load parallel to the web

[29] A_v_min = eta * (h - 2 * t_f) * t_w 6.2.6(3)
= 1 × (203.2 mm - 2 × 11 mm) × 7.2 mm
= 13.046 cm²
Lower bound η h_w t_w

[30] A_v = max(A_v_rolled, A_v_min)
= max(16.944 cm², 13.046 cm²)
= 16.944 cm²
Shear area
[31] N_pl_Rd = A * f_y / gamma_M0 6.2.3(2), 6.2.4(2)
= 58.7 cm² × 355 MPa / 1
= 2083.8 kN
Plastic axial resistance
[32] W_y = W_pl_y 6.2.5(2)
= 497 cm³
= 497 cm³
Section modulus about y (plastic, class 1–2)
[33] W_z = W_pl_z 6.2.5(2)
= 231 cm³
= 231 cm³
Section modulus about z (plastic)
[34] V_pl_Rd = A_v * f_y / (sqrt(3) * gamma_M0) 6.2.6(2)
= 16.944 cm² × 355 MPa / (sqrt(3) × 1)
= 347.28 kN
Plastic shear resistance
[35] u_V = V_Ed / V_pl_Rd
= 20 kN / 347.28 kN
= 0.05759
Shear utilisation
[36] rho = 0 6.2.8(2)
= 0
Shear reduction factor (V_Ed ≤ 0.5 V_pl,Rd)
[37] M_y_c_Rd = (1 - rho) * W_y * f_y / gamma_M0 6.2.5(2), 6.2.8
= (1 - 0) × 497 cm³ × 355 MPa / 1
= 176.44 kN·m
Bending resistance about y
[38] M_z_c_Rd = (1 - rho) * W_z * f_y / gamma_M0 6.2.5(2)
= (1 - 0) × 231 cm³ × 355 MPa / 1
= 82.005 kN·m
Bending resistance about z
[39] u_M_y = M_y_Ed / M_y_c_Rd
= 40 kN·m / 176.44 kN·m
= 0.22671
Bending utilisation, y
[40] u_M_z = M_z_Ed / M_z_c_Rd
= 0 kN·m / 82.005 kN·m
= 0
Bending utilisation, z
[41] u_N = abs(N_Ed) / N_pl_Rd
= abs(-900 kN) / 2083.8 kN
= 0.43189
Compression (cross-section) utilisation
[42] u_section = u_N + u_M_y + u_M_z 6.2.1(7)
= 0.43189 + 0.22671 + 0
= 0.65861
Linear cross-section interaction
[45] alpha_y = imperfection factor, curve b 6.3.1.2
= Designers' Guide §6.3.1
= 0.34
Imperfection factor about y

[46] alpha_z = imperfection factor, curve c 6.3.1.2
= Designers' Guide §6.3.1
= 0.49
Imperfection factor about z
[47] N_cr_y = pi^2 * E * I_y / L_cr_y^2 6.3.1.2(1)
= π^2 × 210 GPa × 4568 cm⁴ / (4 m)^2
= 5917.3 kN
Elastic critical force about y
[48] lambda_y = sqrt(A * f_y / N_cr_y) 6.3.1.2(1)
= sqrt(58.7 cm² × 355 MPa / 5917.3 kN)
= 0.59343
Non-dimensional slenderness about y
[49] Phi_y = 0.5 * (1 + alpha_y * (lambda_y - 0.2) + lambda_y^2) 6.3.1.2(1)
= 0.5 × (1 + 0.34 × (0.59343 - 0.2) + 0.59343^2)
= 0.74296
Φ about y
[50] chi_y = min(1, 1 / (Phi_y + sqrt(Phi_y^2 - lambda_y^2))) 6.3.1.2(1)
= min(1, 1 / (0.74296 + sqrt(0.74296^2 - 0.59343^2)))
= 0.84034
Reduction factor χ about y
[51] N_b_y_Rd = chi_y * A * f_y / gamma_M1 6.3.1.1(3)
= 0.84034 × 58.7 cm² × 355 MPa / 1
= 1751.1 kN
Buckling resistance about y
[52] N_cr_z = pi^2 * E * I_z / L_cr_z^2 6.3.1.2(1)
= π^2 × 210 GPa × 1548 cm⁴ / (4 m)^2
= 2005.3 kN
Elastic critical force about z
[53] lambda_z = sqrt(A * f_y / N_cr_z) 6.3.1.2(1)
= sqrt(58.7 cm² × 355 MPa / 2005.3 kN)
= 1.0194
Non-dimensional slenderness about z
[54] Phi_z = 0.5 * (1 + alpha_z * (lambda_z - 0.2) + lambda_z^2) 6.3.1.2(1)
= 0.5 × (1 + 0.49 × (1.0194 - 0.2) + 1.0194^2)
= 1.2204
Φ about z
[55] chi_z = min(1, 1 / (Phi_z + sqrt(Phi_z^2 - lambda_z^2))) 6.3.1.2(1)
= min(1, 1 / (1.2204 + sqrt(1.2204^2 - 1.0194^2)))
= 0.52876
Reduction factor χ about z
[56] N_b_z_Rd = chi_z * A * f_y / gamma_M1 6.3.1.1(3)
= 0.52876 × 58.7 cm² × 355 MPa / 1
= 1101.9 kN
Buckling resistance about z
[57] N_b_Rd = min(N_b_y_Rd, N_b_z_Rd)
= min(1751.1 kN, 1101.9 kN)
= 1101.9 kN
Governing buckling resistance
[58] u_buckling = abs(N_Ed) / N_b_Rd 6.3.1.1(1)
= abs(-900 kN) / 1101.9 kN
= 0.8168
Flexural buckling utilisation

[59] u_member = abs(N_Ed) / N_b_Rd + k_yy * M_y_Ed / M_y_c_Rd + k_zz * M_z_Ed / M_z_c_Rd 6.3.3(4)
= abs(-900 kN) / 1101.9 kN + 1 × 40 kN·m / 176.44 kN·m + 1 × 0 kN·m / 82.005 kN·m
= 1.0435
Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	16.944 cm²	Shear area
N_pl_Rd	2083.8 kN	N_pl,Rd
V_pl_Rd	347.28 kN	V_pl,Rd
rho	0	ρ
M_y_c_Rd	176.44 kN·m	M_y,c,Rd
M_z_c_Rd	82.005 kN·m	M_z,c,Rd
N_cr_y	5917.3 kN	N_cr,y
N_cr_z	2005.3 kN	N_cr,z
lambda_y	0.59343	λ̄_y
lambda_z	1.0194	λ̄_z
chi_y	0.84034	χ_y
chi_z	0.52876	χ_z
N_b_y_Rd	1751.1 kN	N_b,y,Rd
N_b_z_Rd	1101.9 kN	N_b,z,Rd
N_b_Rd	1101.9 kN	N_b,Rd
u_N	0.43189	Axial utilisation
u_M_y	0.22671	Bending y utilisation
u_M_z	0	Bending z utilisation
u_V	0.05759	Shear utilisation
u_section	0.65861	Cross-section interaction
u_buckling	0.8168	Buckling utilisation
u_member	1.0435	Member interaction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Member interaction (simplified)	u_member = 1.0435	u_member ≤ 1	1.04	6.3.3
PASS	Flexural buckling N_Ed ≤ N_b,Rd	u_buckling = 0.8168	u_buckling ≤ 1	0.817	6.3.1.1
PASS	Cross-section interaction	u_section = 0.65861	u_section ≤ 1	0.659	6.2.1(7)
PASS	Compression N_Ed ≤ N_c,Rd	u_N = 0.43189	u_N ≤ 1	0.432	6.2.4

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bending y M_y,Ed ≤ M_y,c,Rd	u_M_y = 0.22671	u_M_y ≤ 1	0.227	6.2.5
PASS	Shear V_Ed ≤ V_pl,Rd	u_V = 0.05759	u_V ≤ 1	0.0576	6.2.6
PASS	Bending z M_z,Ed ≤ M_z,c,Rd	u_M_z = 0	u_M_z ≤ 1	0	6.2.5

7. Applicability

WARNING — no-ltb: M_y_Ed: lateral-torsional buckling is not checked by this sheet; restrain the compression flange or check 6.3.2 separately.

8. Verification

Template struct.en1993-member@1.0.1 Content hash 7e45c53b9cd8fa25 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0